Supercritical Inverse-Square Dynamics: Domains, Collisions and the Complete Bound Ladder

HCS-C391 theorem and reproducibility package

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Abstract

We reconstruct the complete self-adjoint boundary family of the strictly supercritical inverse-square Hamiltonian on the positive half-line. Its classical counterpart has an exact quadratic position law, and every trajectory reaches an excluded collision endpoint in at least one time direction. Endpoint analysis gives a circle of quantum boundary conditions, none selected by the classical differential expression. The Bessel Green function determines every simple negative level and its normalized eigenfunction. The levels form a bilateral geometric sequence with ratio $\exp(-2\pi/\sigma)$, accumulating at zero in one direction and diverging to minus infinity in the other. In particular, no self-adjoint realization has a ground state or a Friedrichs construction. The normalization is derived from a two-momentum Lagrange identity, and branch changes only relabel the levels. The formulas retain their classical literature ownership. This first draft closes the source clock, all boundary domains and the complete bound ladder; finite rational and special-function checks are distinguished from proof.

Keywords: inverse-square Hamiltonian; self-adjoint domain; bilateral bound spectrum; continuous scattering; discrete scaling; determinant obstruction.

中文摘要：
本文完整整理半直线超临界反平方势的自伴边界族。经典位置平方满足精确二次律，
每条轨道至少在一个时间方向抵达被排除的碰撞端点。量子边界构成一个圆，
并不由经典方程唯一选定。贝塞尔核给出全部简单负本征值及归一化函数。
双向几何能级分别趋于零和负无穷，因此没有基态或弗里德希斯实现。
本文保留经典文献归属，有限核验不替代完整证明。
关键词：反平方势；自伴定义域；双向负谱；连续散射；离散尺度；行列式障碍

1 A local Hamiltonian does not choose a boundary

Fix $\sigma > 0$, put $g = \sigma^2 + 1/4$, and use units $\hbar = 2m = 1$. The classical source is $h(x, p) = p^2 - g/x^2$ on $T^*(0, \infty)$. Its quantum expression is $\ell = -\partial_x^2 - g/x^2$ on $\mathcal{H} = L^2((0, \infty), \mathrm{d}x)$. A unit complex number κ will specify the missing boundary condition. It is independent physical input, not a value fitted to a desired eigenvalue or prime scale. Throughout, I_ν, J_ν, K_ν are ordinary Bessel functions.

The contribution is a single-convention source theorem and audit, not a new discovery of the inverse-square spectrum. Dereziński and Richard give the domain, spectrum and scattering theory in a substantially larger complex parameter family [1]. We retain the entire strictly supercritical self-adjoint subfamily, derive the constants used here, and connect them to the original source clock and the limits of ordinary global determinants. The critical value $\sigma = 0$, nonunit κ, ultraviolet regularization and confinement are not part of this theorem.

Proposition 1 (The complete collision clock). *For every initial point (x_0, p_0), with energy $E = h(x_0, p_0)$,*

$$y(t) = x(t)^2 = x_0^2 + 4x_0p_0t + 4Et^2, \qquad p(t) = \frac{y'(t)}{4\sqrt{y(t)}}. \tag{1}$$

Its maximal time interval is the component of $\{t : y(t) > 0\}$ containing zero. Every orbit has a finite collision endpoint, and none is periodic in the declared phase space.

Proof. Hamilton's equations are $x' = 2p$ and $p' = -2g/x^3$, so $y'' = 8E$. The reconstruction in (1) satisfies both equations. For $E \neq 0$ the quadratic discriminant is $16g > 0$. At $E < 0$ its positive component is bounded; at $E > 0$ it is a half-line. At $E = 0$ the nonzero slope is $\pm 4\sqrt{g}$, again giving a half-line. Each finite endpoint has $x = 0$, which is excluded. No collision continuation has been added, so none of these intervals gives a periodic orbit. □

2 All self-adjoint domains and the bound ladder

Let $D_{\max} = \{f \in L^2 : \ell f \in L^2$ distributionally$\}$ and let $D_{\min}$ be the graph closure of $C_c^\infty(0,\infty)$. Membership in $D_{\min}$ near zero means membership after multiplication by a smooth cutoff equal to one there. Define

$$D_\kappa = \{f \in D_{\max} : f - c(\kappa x^{1/2-\mathrm{i}\sigma} + x^{1/2+\mathrm{i}\sigma}) \in D_{\min} \text{ near } 0 \text{ for some } c \in \mathbb{C}\}. \tag{2}$$

Theorem 1 (Exhaustive boundary family). *The domains (2), $|\kappa| = 1$, are exactly the self-adjoint extensions of the minimal operator. None is semibounded, and there is no Friedrichs realization. Position-space complex conjugation preserves every one of these domains.*

Proof. The zero-energy basis $u_\pm = x^{1/2\pm\mathrm{i}\sigma}$ is square integrable near zero. Variation of constants gives a unique expansion $f = a_f u_- + b_f u_+$ modulo $D_{\min}$ there. Indeed, the integrals of $u_\pm \ell f$ converge by Cauchy–Schwarz and their remainders are $o(x^{3/2})$, with derivatives $o(x^{1/2})$. Infinity is limit point since the potential is integrable on tails. Weyl's endpoint theorem therefore gives deficiency indices $(1, 1)$. The boundary Wronskian is

$$W_0(\bar{f}, v) = 2\mathrm{i}\sigma(\overline{b_f} b_v - \overline{a_f} a_v).$$

Its maximal isotropic lines are exactly $a = \kappa b$, $|\kappa| = 1$; neither coefficient of a nonzero isotropic vector can vanish. Conjugation changes the ratio to $1/\overline{\kappa} = \kappa$. For nonsemiboundedness, set $f(x) = x^{1/2}\chi(\log x)$ with a real compactly supported smooth function χ. Direct substitution gives

$$\langle f, \ell f\rangle = \int_{\mathbb{R}} (|\chi'(s)|^2 - \sigma^2|\chi(s)|^2)\,\mathrm{d}s.$$

A sufficiently long plateau makes this negative. Normalized dilations make it arbitrarily negative. Every extension contains these minimal-domain test functions, so no semibounded extension exists. □

Put $\varsigma = \kappa\Gamma(-\mathrm{i}\sigma)/\Gamma(\mathrm{i}\sigma) = \mathrm{e}^{\mathrm{i}\theta}$, using any real lift θ. Powers of a right-half-plane momentum use the logarithm on $\operatorname{Re}\rho > 0$.

Theorem 2 (The entire negative spectrum). *Every negative spectral point of $H_{\sigma,\kappa} = \ell|_{D_\kappa}$ is a simple eigenvalue*

$$E_j = -\rho_j^2 = -4\exp\left(-\frac{\theta + 2\pi j}{\sigma}\right), \quad j \in \mathbb{Z}, \qquad \psi_j(x) = \rho_j\sqrt{\frac{2\sinh(\pi\sigma)}{\pi\sigma}}\sqrt{x}K_{\mathrm{i}\sigma}(\rho_j x). \tag{3}$$

The functions ψ_j have unit norm. A different lift of θ merely reindexes the same spectrum.

Proof. At energy $-\rho^2$ the decaying solution at infinity is uniquely $\sqrt{x}K_{\mathrm{i}\sigma}(\rho x)$, up to a scalar. Its endpoint coefficients follow from

$$K_{\mathrm{i}\sigma}(w) = \tfrac{1}{2}\Gamma(\mathrm{i}\sigma)(w/2)^{-\mathrm{i}\sigma} + \tfrac{1}{2}\Gamma(-\mathrm{i}\sigma)(w/2)^{\mathrm{i}\sigma} + o(1).$$

Their ratio equals κ exactly when $\varsigma(\rho/2)^{2\mathrm{i}\sigma} = 1$, giving (3). For $\operatorname{Re}\rho > 0$ this equality forces $\arg\rho = 0$ by taking absolute values. Away from these roots the Green kernel in the proof supplement gives a bounded inverse; hence no additional negative spectrum is omitted. Uniqueness of the decaying solution gives simplicity.

For $f_a(x) = \sqrt{x}K_{\mathrm{i}\sigma}(ax)$ and $f_b(x) = \sqrt{x}K_{\mathrm{i}\sigma}(bx)$, the Lagrange identity is $W(f_a, f_b)' = (b^2 - a^2)f_a f_b$. The Wronskian vanishes at infinity; its endpoint value is $\sigma|\Gamma(\mathrm{i}\sigma)|^2\sin(\sigma\log(a/b))$. Using $|\Gamma(\mathrm{i}\sigma)|^2 = \pi/(\sigma\sinh\pi\sigma)$ gives

$$\int_0^\infty xK_{\mathrm{i}\sigma}(ax)K_{\mathrm{i}\sigma}(bx)\,\mathrm{d}x = \frac{\pi\sin(\sigma\log(a/b))}{(a^2 - b^2)\sinh(\pi\sigma)}. \tag{4}$$

The coincident limit is $\pi\sigma/(2a^2\sinh\pi\sigma)$, proving the normalizer in (3). Endpoint $O(x)$ bounds and exponential tail decay justify that limit. Distinct ladder values give orthogonality in (4). Finally $E_j \to 0$ as $j \to +\infty$ and $E_j \to -\infty$ as $j \to -\infty$; there is no lowest level. □

3 Reproducibility and literature ownership

The exact producer contains 45 classical phase points, 27 boundary-flux rows and 15 algebraic scattering rows. A non-importing SymPy checker reconstructs their values and literal scalar types. Sixty bound-level and 36 continuum rows are stored to 60 digits after 100-digit evaluation and independently recomputed at 110 digits through endpoint-normalized coefficients. The separate symbolic lane checks eight identities. These finite grids are regression, not interval certification or a proof of spectral completeness. Two unrelated working-directory replays, repaired-hash attacks, strict YAML gates and smoke tests are distributed with the proof. The release rebuilds all three drafts in fresh directories with frozen epoch 1788566400 and retains actual settled compiler logs.

The external mathematical ownership is concentrated in one directly inspected primary source [1]; listing its journal and preprint as two independent works would be misleading. The paper does not claim priority for the self-adjoint family, geometric ladder or scattering formulas. Its package-level advance is the joined, auditable boundary between that complete source theorem and a target spectral construction it does not provide. The neighboring repository models are a Friedrichs Aharonov–Bohm Hamiltonian, a repulsive many-body Calogero–Moser system and a positive isotonic oscillator. They do not contain this strictly supercritical, imaginary-order, all-domain bilateral ladder. The proof and source audit preserve those owner differences.

Local collaborating AI agents reviewed the proof and its normalization; this is not external human peer review or cross-provider validation. An independent reading prompted direct Green/Stone tests and a fully expanded normalization identity, both implemented here. No outside model was invoked.

4 Conclusion

Round zero: domains and the complete bound ladder. The original source has an explicit collision clock and a complete circle of quantum self-adjoint domains. The bilateral negative spectrum has no lowest level. This closes the endpoint and bound-state foundation without pretending that a classical collision law selected a boundary phase.

References

[1] J. Dereziński and S. Richard. On Schrödinger operators with inverse square potentials on the half-line. *Annales Henri Poincaré* **18**, 869–928 (2017). doi:10.1007/s00023-016-0520-7. Author manuscript: arXiv:1604.03340v2. The theorem numbering used here follows that inspected preprint.